

**TAREA T3: CONTINUACIÓN DE CONVERSIONES Y
OPERACIONES ARITMÉTICAS DE SUMA Y MULTIPLICACIÓN**

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I) Realizar las conversiones siguientes en base potencia 2:

① $125764.567_8 \rightarrow X_2$

<u>1</u>	<u>2</u>	<u>5</u>	<u>7</u>	<u>6</u>	<u>4</u>	<u>5</u>	<u>6</u>	<u>7</u>
001	010	101	111	110	100	101	110	111

$125764.567_8 \rightarrow 101010111110100.101110111_2$

② $1101101011.0110011_2 \rightarrow X_4 \rightarrow Y_8 \rightarrow Z_{16}$

$\rightarrow X_4$

<u>1</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>
3	1	2	2	3	1	2	1	2	1	2										

$1101101011.0110011_2 \rightarrow 31223.1212_4$

$\star Y_8$

<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>
1	5	5	3	3	1	4														

$1101101011.0110011_2 \rightarrow 1553.314_8$

00110110101101100110

3 6 B 6 6

1101101011.0110011₂ → 36B,66₁₆

③ 2230213,0321₄ → X₈ → Y₁₆ → Z₃₂

2 2 3 0 2 1 3 . 0 3 2 1

010 10 11 00 10 01 11 . 00 11 10 010

2 5 4 4 7 1 6 2

2230213,0321₄ → 25447.162₈

* Y₁₆

2 2 3 0 2 1 3 . 0 3 2 1

0010 10 11 00 10 01 11 . 00 11 10 01

2 B 2 7 3 9

2230213.0321₄ → 2B2739₁₆

* Z₃₂

2 2 3 0 2 1 3 . 0 3 2 1

010 10 11 00 10 01 11 . 00 11 10 0100

A P 7 . 7 4

2230213.0321 → AP7.74

④ A0FC4A.2CE5₁₆ → X₂ → Y₄ → Z₈

A	0	F	C	4	A	.	2	C	E	5
1010	0000	1111	1100	0100	1010	.	0010	1100	1110	0101

A0FC4A.2CE5₁₆ → 101000001111110001001010.0010110011100101₂

* Y₄

A	0	F	C	4	A	.	2	C	E	5
1010	0000	1111	1100	0100	1010	.	0010	1100	1110	0101

2	2	0	0	3	3	3	0	1	0	2	2	.	0	2	3	0	3	2	1	1
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A0FC4A.2CE5 → 220033301022,02303211₄

* Z₈

A	0	F	C	4	A	.	2	C	E	5	
1010	0000	1111	1100	0100	1010	.	0010	1100	1110	0101	06

5	0	1	7	6	1	1	2	.	1	3	1	6	2	4
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A0FC4A.2CE5 → 50176112.131624₈

5. 53G78.H604₃₂ → Y₄ → Z₈ → V₁₆

* Y₄

5 3 G 7 8 . H 6 0 4
0010100011 10000 00111 01000 10001 00110 00000 00100
110130003220.2021200010

53G78.H604₃₂ → 110130003220.2021200010₄

* Z₈

5 3 G 7 8 . H 6 0 4
000101 00011 10000 00111 01000 10001 00110 00000 001000
24340350.4230010

53G78.H604₃₂ → 24340350.4230010₈

* V₁₆

5 3 G 7 8 . H 6 0 4
00101 00011 10000 00111 01000 10001 00110 00000 00100
51C0E8.89804

53G78.H604₃₂ → 51C0E8.89804₁₆

II) Realizar las tablas de suma y multiplicación de las siguientes bases:

1) Base 5:

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	10
2	2	3	4	10	11
3	3	4	10	11	12
4	4	10	11	12	13

x	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	11	13
3	0	3	11	14	22
4	0	4	13	22	31

2) Base 8:

+	0	1	2	3	4	5	6	7
0	0	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7	10
2	2	3	4	5	6	7	10	11
3	3	4	5	6	7	10	11	12
4	4	5	6	7	10	11	12	13
5	5	6	7	10	11	12	13	14
6	6	7	10	11	12	13	14	15
7	7	10	11	12	13	14	15	16

x	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7
2	0	2	4	6	10	12	14	16
3	0	3	6	11	14	17	22	25
4	0	4	10	14	20	24	30	34
5	0	5	12	17	24	31	36	43
6	0	6	14	22	30	36	44	52
7	0	7	16	25	34	43	52	61

3)Base 16:

+	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
1	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	11
2	2	3	4	5	6	7	8	9	A	B	C	D	E	F	10	11
3	3	4	5	6	7	8	9	A	B	C	D	E	F	10	11	12
4	4	5	6	7	8	9	A	B	C	D	E	F	10	11	12	13
5	5	6	7	8	9	A	B	C	D	E	F	10	11	12	13	14
6	6	7	8	9	A	B	C	D	E	F	10	11	12	13	14	15
7	7	8	9	A	B	C	D	E	F	10	11	12	13	14	15	16
8	8	9	A	B	C	D	E	F	10	11	12	13	14	15	16	17
9	9	A	B	C	D	E	F	10	11	12	13	14	15	16	17	18
A	A	B	C	D	E	F	10	11	12	13	14	15	16	17	18	19
B	B	C	D	E	F	10	11	12	13	14	15	16	17	18	19	1A
C	C	D	E	F	10	11	12	13	14	15	16	17	18	19	1A	1B
D	D	E	F	10	11	12	13	14	15	16	17	18	19	1A	1B	1C
E	E	F	10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D
F	F	10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D	1E

x	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
2	0	2	4	6	8	A	C	E	10	12	14	16	18	1A	1C	1E
3	0	3	6	9	C	F	12	15	18	1B	1E	21	24	27	2A	2D
4	0	4	8	C	10	14	18	1C	20	24	28	2C	30	34	38	3C
5	0	5	A	F	14	19	1E	23	28	2D	32	37	3C	41	46	4B
6	0	6	C	12	18	1E	24	2A	30	36	3C	42	48	4E	54	5A
7	0	7	E	15	1C	23	2A	31	38	3F	46	4D	54	5B	62	69
8	0	8	10	18	20	28	30	38	40	48	50	58	60	68	70	78
9	0	9	12	1B	24	2D	36	3F	48	51	5A	63	6C	75	7E	87
A	0	A	14	1E	28	32	3C	46	50	5A	64	6E	78	82	8C	96
B	0	B	16	21	2C	37	42	4D	58	63	6E	79	84	8F	9A	A5
C	0	C	18	24	30	3C	48	54	60	6C	78	84	90	9C	A8	B4
D	0	D	1A	27	34	41	4E	5B	68	75	82	8F	9C	A9	B6	C3
E	0	E	1C	2A	38	46	54	62	70	7E	8C	9A	A8	B6	C4	D2
F	0	F	1E	2D	3C	4B	5A	69	78	87	96	B0	C0	D0	E0	F0

III) Realizar las operaciones siguientes:

1.- $4735.26 + 61427.55 + 2034.57 + 2765.4351$ Base 8

$$\begin{array}{r} 4735.26 \\ + 61427.55 \\ + 2034.57 \\ + 2765.4351 \\ \hline 73407.2551 \end{array} \quad (\text{Base 8})$$

2.- $3434.24 + 2042.102 + 1234 + 321.4321$ Base 5

$$\begin{array}{r} 3434.2400 \\ + 2042.1020 \\ + 1234.0000 \\ + 321.4321 \\ \hline 13142.3241 \end{array} \quad (\text{Base 5})$$

3.- $5235.31 + 2304.1 + 4234.531 + 345.5301$ Base 6

$$\begin{array}{r}
 5235.310 \\
 + 2304.100 \\
 4234.531 \\
 \underline{345.5301} \\
 21012.3111 \quad (\text{Base } 6)
 \end{array}$$

4.- $9E75.6 + BA4CF.FE1 + 248F.057 + 7C6A5.4D$ Base 16

$$\begin{array}{r}
 9E75.6 \\
 + BA4CF.FE1 \\
 248F.057 \\
 \underline{7C6A5.4D} \\
 142E71.808 \quad (\text{Base } 16)
 \end{array}$$

5.- $233223.32 + 2030.1$ Base 4

$$\begin{array}{r}
 233223.32 \\
 \times \quad \underline{2030.1} \\
 23322332 \\
 + \quad 00000000 \\
 203300322 \\
 000000000 \\
 \underline{113311330} \\
 1220133321132 \quad (\text{Base } 4)
 \end{array}$$

6.- $369C.F02 \times 974A.BD$

Base 16

$$\begin{array}{r}
 \times \quad 369C.F02 \\
 \quad 974A.BD \\
 \hline
 \quad 2C5F831A \\
 \quad 258BE516 \\
 + \quad 22221614 \\
 \quad DA73C08 \\
 \quad 17E4A90E \\
 \hline
 1E684712 \\
 \hline
 20468352.2687A \quad (\text{Base } 16)
 \end{array}$$

7.- 3726.304×35.3401

Base 8

$$\begin{array}{r}
 \times \quad 3726.304 \\
 \quad 35.3401 \\
 \hline
 \quad 3726304 \\
 \quad 0000000 \\
 + \quad 17531420 \\
 \quad 13603114 \\
 \quad 23457724 \\
 \hline
 13603114 \\
 \hline
 163267,3044304 \quad (\text{Base } 8)
 \end{array}$$

8.- 33021.232×3023.31

Base 4

$$\begin{array}{r}
 \times \quad 33021.232 \\
 \quad 3023.31 \\
 \hline
 \quad 33021232 \\
 \quad 231131012 \\
 + \quad 231131022 \\
 \quad 132103130 \\
 \quad 00000000 \\
 \hline
 231131022 \\
 \hline
 300100020.30312 \quad (\text{Base } 4)
 \end{array}$$